

Databases for Big Data

Exercises: Query optimization

I.Savnik, 10/1/2026.

Example database Sales

The following database contains two entity tables, Customer and Product, which store descriptive information about business entities. A Sales relation links these entities and records measurable events (e.g., quantities sold), forming a simple OLAP schema suitable for analytical queries.

```
CREATE TABLE Customer (  
  cid    INT PRIMARY KEY,  
  name   TEXT NOT NULL,  
  region TEXT  
);
```

```
CREATE TABLE Product (  
  pid     INT PRIMARY KEY,  
  name    TEXT NOT NULL,  
  category TEXT  
);
```

```
CREATE TABLE Sales (  
  sale_id INT PRIMARY KEY,  
  cid     INT NOT NULL,  
  pid     INT NOT NULL,  
  quantity INT NOT NULL,  
  FOREIGN KEY (cid) REFERENCES Customer(cid),  
  FOREIGN KEY (pid) REFERENCES Product(pid)  
);
```

Here are the examples of the tables that are instances of the above relations.

Product

pid	name	category
1	Laptop	Electronics
2	Desk Chair	Furniture
3	Coffee Maker	Appliances
4	Notebook	Stationery
5	Headphones	Electronics
6	Desk Lamp	Furniture
7	Water Bottle	Accessories

Customer

cid	name	region
1	Alice Brown	West
2	Bob Smith	East
3	Carla Torres	North
4	Daniel Kim	South
5	Eva Johnson	Central

Sales

sale_id	cid	pid	quantity
1	1	1	2
2	1	4	5
3	2	3	1
4	2	7	3
5	3	2	2
6	3	5	1
7	4	1	1
8	4	6	2
9	5	3	4
10	5	7	2

Queries

Query 1.

```
select P.pid, P.category
from Product P, Sales S
where P.name = 'Notebook'
and    P.pid=S.pid
and    P.name='Laptop'
and    S.quantity<4;
```

Query 2.

```
select C.name, P.name, S.quantity
from Product P, Sales S, Customer C
where C.cid=S.cid
and    P.pid=S.pid
and    C.region='West'
and    P.category='Electronics'
and    S.quantity>5;
```

Exercises

Exercise 1.

Demonstrate how the heuristic-based optimization algorithm from the early relational system Ingres would optimize Query 1 and Query 2. Show the final query plans!

Exercise 2.

Sketch the execution of the System R query optimizer using dynamic programming on Query 1 and Query 2. Present the final optimized query plans!

Exercise 3.

- a) Describe the main concepts of the Starburst query optimizer.
- b) Sketch the execution of the Starburst query optimizer applied to Query 1 and Query 2.

Exercise 4.

- a) Give an abstract algorithmic description of the Volcano optimizer.
- b) Sketch the evaluation of the Volcano optimizer on Query 1 and Query 2.

Exercise 5.

- a) Present the main concepts and the algorithm used in the Cascades optimizer.
- b) Sketch the evaluation of the Cascades optimizer on Query 1 and Query 2.